

Foundation Models for Classification of Soft Tissue Tumors

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Abstract. This research evaluates the characteristics of the SAM-Med3D base model for small sample size classification of tumors on six clinical data sets (CRLM, desmoids, GIST, liposarcomas, liver, melanomas). Volumetric scans from CT/MRI were resized to 128x128x128 and then cropped based on region-of-interest (ROI)-centric cropping or full volume resizing and optionally filtered by lesion size to exclude poor-quality samples. Three-dimensional encoder features were extracted and compared with different pooling strategies using percentile pooling as the default method to produce 1920-dimensional case embeddings followed by standardization to the training partition and principal component analysis. Classification was subsequently performed downstream with TabPFN and LoCalPFN, allowing in-context inference without requiring end-to-end re-training. Experiments evaluated the SAM-Med3D feature pipeline against two three-dimensional baselines (3D DenseNet-121 and 3D ViT), including an evaluation of attaching a light-weight classification head to a frozen SAM-Med3D encoder to evaluate feature quality directly. The results highlight current limitations of SAM-Med3D as a feature-extractor for tumor classification in clinically relevant work-flow.

Keywords: First keyword · Second keyword · Another keyword.

1 Introduction

Soft tissue tumors form a heterogeneous group of lesions that arise from mesenchymal tissues and present substantial diagnostic and therapeutic challenges [5]. Accurate non-invasive diagnosis is essential for treatment planning and prognosis, but tumors often exhibit similar characteristics on scans, and definitive confirmation frequently requires invasive testing that may not be possible before therapy. Medical imaging advancements in combination with machine learning technology enable better diagnostic results, but their clinical application faces three main barriers: insufficient training data, high processing requirements for 3D medical images and the need for universal features that maintain stability [11, ?]. The training of foundation models on extensive medical volume datasets

presents a potential solution to these problems because they generate adaptable embeddings which need less labeling information while reducing training and inference expenses and enhancing performance between different healthcare facilities [16, ?].

The proposed pipeline leverages large-scale pretrained volumetric representations together with compact, data-efficient classifiers. The approach reuses semantic features learned by a foundation model for 3D medical imaging to reduce the amount of task-specific labeled data required, while a lesion-centric preprocessing strategy and region-of-interest pooling concentrate computational effort on the most informative parts of the scan. The extracted embeddings are standardized and optionally reduced in dimensionality. They are then provided to transformer-based tabular classifiers [6, ?], which are optimized for small structured feature sets and enable rapid model testing without full volumetric training.

The implemented system employs SAM-Med3D [16] as a feature extractor to obtain rich embeddings from regions containing lesion, applies adaptive ROI pooling and lesion filtering to produce compact feature vectors per study, and uses TabPFN [6] and LoCalPFN [15] as classifiers. Complementary experiments compare these transfer-learning workflows to native 3D classification heads trained on cropped volumes, allowing evaluation of the trade-offs between feature reuse and end-to-end learning under matched preprocessing conditions.

Experimental data comprise 930 CT and MRI studies from the publicly available WORC dataset [14], spanning multiple tumor categories including gastrointestinal stromal tumors, lipomatous tumors, desmoid-type fibromatosis, liver lesions, and melanoma metastases.

The research evaluates whether pre-trained 3D embeddings, combined with transformer-based tabular classifiers and limited labeled data, can achieve competitive performance on medical classification tasks. It further compares alternative data-preparation strategies for volumetric lesions and develops a systematic framework for running experiments across multiple datasets and aggregating their results.[12, ?].

2 Data

2.1 The WORC Database

The imaging data used in this study originates from the WORC database, a publicly available collection of CT and MR scans with corresponding segmentations and clinical labels assembled specifically for radiomics research on soft tissue tumors [14, ?]. The database comprises 930 patients from six independent radiomics studies conducted at the Erasmus MC, making it the largest publicly available resource for soft tissue tumor classification research. Each case includes volumetric imaging data in NIfTI format, expert-delineated binary segmentation masks identifying one or more lesion regions, and clinical labels encoding diagnostic outcomes or tumor subtypes.

The database encompasses six distinct tumor categories, each representing a clinically relevant binary classification task. Gastrointestinal stromal tumors (GIST) comprise 246 cases with the classification objective of distinguishing mutation status or malignancy grade. Lipomatous tumors include 115 cases where the task is to differentiate benign lipomas from liposarcomas, a distinction with significant treatment implications. Desmoid-type fibromatosis contains 203 cases focused on predicting tumor progression or stability. Liver lesions encompass 186 cases addressing the differentiation of hepatocellular carcinoma from other liver masses. Colorectal liver metastases (CRLM) include 77 cases with classification targets related to treatment response or resectability. Melanoma metastases comprise 103 cases where the objective is prediction of immunotherapy response or lesion characterization.

Table 1. Summary of the WORC database tumor categories used in this study.

Category	GIST	Lipo	Desmoid	Liver	CRLM	Melanoma	Total
Cases	246	115	203	186	77	103	930
Modality	CT	MRI	MRI	CT/MRI	CT	CT	–

2.2 Why WORC: A Challenging Benchmark

The WORC database served as the research choice because it presents one of the most difficult medical image classification benchmarks which researchers can access freely. The dataset contains specific features which make it challenging to work with, but more importantly, it is closer to real-world clinical tasks where foundation models are not adequately evaluated, thus making it an appropriate environment to assess foundation model performance in realistic medical applications.

The classification tasks involve subtle imaging differences that are difficult even for expert radiologists to distinguish without histopathological confirmation. The process of identifying benign lipomas from well-differentiated liposarcomas demands pathologists to identify small details which appear identical between these two tumor types [9]. The evaluation of GIST tumor mutation status through imaging data represents the maximum extent which visual characteristics can disclose about molecular tumor properties [13]. If foundation models can achieve reasonable performance on these ambiguous tasks, they are likely to generalize to a wide range of clinical applications where the imaging signal is more pronounced.

The database shows wide variation in imaging methods and scanner brands and scanning methods which affect the six different tumor types. The different characteristics in this data set represent what clinicians experience during their work because they cannot follow standardized procedures which creates difficulties for machine learning models to learn from protocol-based data. A model

that achieves high performance in all six categories of WORC demonstrates the stability required for real-world deployment.

The study contains small tumor category sample sizes, including between 77 and 246 cases for each category. While traditional deep learning approaches require large datasets, pre-trained foundation models are particularly well-suited for this data-scarce scenario, as they leverage knowledge from large-scale pre-training to perform effectively with limited task-specific data. The majority of medical imaging applications encounter this limitation because rare diseases and specialized centers and expensive expert annotation services limit the amount of available labeled data. Methods that require thousands of training examples will fail in this regime, making WORC an ideal benchmark for evaluating data-efficient approaches such as foundation model transfer learning combined with small-data classifiers.

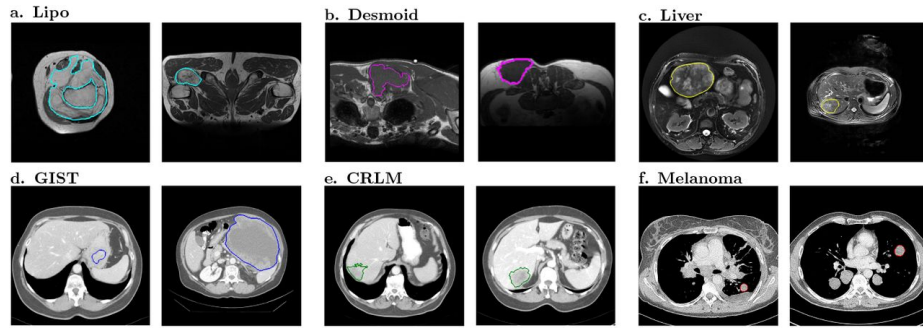


Fig. 1. Example CT and MRI slices from the WORC database showing the diversity of tumor appearances and imaging modalities. Top row: GIST and CRLM (CT). Bottom row: lipomatous tumor and desmoid (MRI).

2.3 Implications for Foundation Model Evaluation

The combination of task difficulty, data heterogeneity, sample size constraints, and strong baselines makes WORC an ideal proving ground for foundation model approaches to medical image classification. The central hypothesis of this thesis is that pretrained volumetric embeddings from SAM-Med3D, combined with data-efficient tabular classifiers, can achieve competitive performance on these challenging tasks without task-specific training of the image encoder.

The research suggests that foundation models trained on extensive medical imaging data learn representations which successfully apply to subsequent classification tasks that require identifying small visual differences across various imaging methods with restricted available data. The discovery would have significant implications for medical image analysis and AI, as it would decrease the amount of data and computational power needed to develop diagnostic tools

from medical images, potentially accelerating the translation of machine learning research toward clinical applications.

On the contrary, if foundation model transfer learning fails to match optimized radiomics baselines on WORC, it would highlight important limitations of current pretrained models and suggest directions for improving medical imaging foundation models. The results from both scenarios will generate vital information which scientists need to establish the correct operational parameters for foundation model medical image analysis.

3 Methods

The workflow consists of four main stages: data preparation and quality control, feature extraction from volumetric scans using pretrained models, dimensionality reduction and standardization, and classification with tabular foundation models designed for small datasets.

The first stage, data preparation, involves organizing imaging data and meta-data from the WORC dataset [14], a single-center collection comprising multiple tumor categories. The data set contains CT and magnetic resonance volumes that include one or more regions of the lesion that need annotation and their corresponding diagnostic labels. The research team tested different preprocessing methods to find the best combination that would maintain important imaging data while keeping processing time efficient. The evaluation included three methods, which were full-volume processing to maintain complete scan data and lesion size filtering through voxel count and morphological criteria to remove small or poor-quality annotations and lesion-centered region-of-interest cropping with adaptive padding to extract standardized tumor volumes [3]. The research applied region-of-interest cropping as its main preprocessing technique to meet SAM-Med3D’s fixed input requirement of $128 \times 128 \times 128$ voxels. This approach focuses computational resources on tumor-related features while standardizing all studies to the required input size.

The second stage of processing extracts high-level semantic features from the preprocessed volumes. The analysis uses SAM-Med3D [16] as its foundation model because it was trained on multiple medical imaging tasks instead of starting from scratch with a volumetric convolutional network. The three-dimensional input of SAM-Med3D gets processed to generate dense embeddings which get distributed across all spatial locations. The model uses these embeddings to create a single feature vector for each study through a percentile pooling layer which combines both appearance and spatial information about the lesion [16, ?].

The third stage of the process prepares the extracted feature vectors for classification. The features are standardized to zero mean and unit variance, and principal component analysis is applied to reduce dimensionality while retaining the directions of greatest variance [8]. The method reduces model overfitting risks while speeding up the training process for future classifiers. For studies with multiple lesions, features are pooled at the patient level according to a predefined aggregation scheme to produce one feature vector per case.

The fourth stage performs binary classification using transformer-based tabular models, specifically TabPFN [6] and LoCalPFN [15], which are designed to excel with limited training data by incorporating strong inductive biases and performing in-context learning. All experiments follow a stratified splitting protocol that preserves class balance in both training and validation sets, and performance is evaluated using accuracy, area under the receiver operating characteristic curve, sensitivity, specificity, and F1 score. Cross-validation or repeated holdout validation is used where sample sizes permit to obtain confidence estimates for each metric.

3.1 Data Preparation and Preprocessing

Several preprocessing strategies were evaluated to determine the optimal trade-off between preserving clinically relevant imaging information and maintaining computational efficiency. Three candidate approaches were considered. Full-volume processing to retain the entire scan without modification before passing it to the encoder; this preserves maximal anatomical context but results in variable input sizes and includes large regions of non-lesion tissue that may introduce noise. Lesion size filtering to apply voxel-count and morphological criteria to exclude annotations that are too small or irregularly shaped to yield reliable features; this improves signal quality but discards a subset of cases and does not address input size variability. Lesion-centered region-of-interest cropping with adaptive padding to extract a fixed-size subvolume centered on the lesion bounding box and pads or crops as needed to produce uniform input dimensions across all studies [3].

SAM-Med3D requires a fixed input size of $128 \times 128 \times 128$ voxels, which can be achieved through various preprocessing strategies. Comparative experiments indicated that region-of-interest cropping offered the best approach to meet this requirement for the objectives of this work. Centering the crop on the tumor ensures that the encoder receives input dominated by lesion tissue and its immediate surroundings, reducing the influence of distant anatomy and scanner-specific field-of-view variations. This constraint simplifies batching during feature extraction and ensures that all cases contribute equally to downstream analyses regardless of original scan dimensions. The adaptive padding scheme handles cases where the lesion is located near the edge of the scan by reflecting or zero-padding the boundary, avoiding the need to discard edge cases. Based on these considerations, region-of-interest cropping was adopted as the primary preprocessing method for all subsequent experiments, with lesion size filtering applied as an optional quality-control step when stricter inclusion criteria were desired.

3.2 SAM-Med3D: Foundation Model for Volumetric Medical Imaging

SAM-Med3D is a foundation model designed specifically for three-dimensional medical image segmentation and feature extraction [16]. The model extends the Segment Anything Model architecture from natural 2D images to volumetric

medical data by replacing all 2D operations with their 3D counterparts and adapting the positional encoding to handle three spatial dimensions. The system contains three main components which include a Vision Transformer image encoder that extracts dense embeddings from the input volume and a prompt encoder that processes user-defined guidance through point clicks and bounding boxes and a mask decoder that produces segmentation output. For the purposes of this work, only the image encoder is used to extract feature representations, discarding the prompt encoder and mask decoder components.

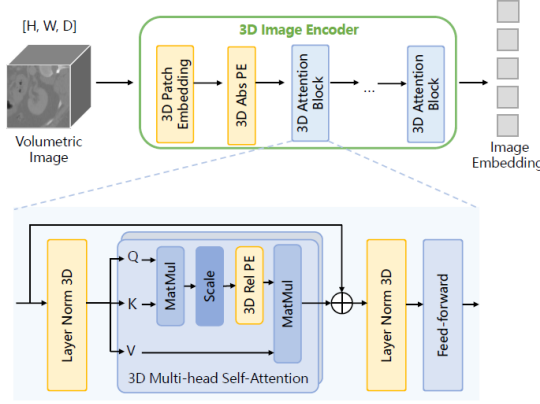


Fig. 2. Architecture of SAM-Med3D. The model consists of a 3D Vision Transformer image encoder, a prompt encoder for interactive guidance, and a mask decoder. For feature extraction, only the image encoder is used.

The image encoder employs a ViT-B backbone that divides the input volume into non-overlapping 3D patches, projects each patch into a high-dimensional embedding space, and processes the sequence of patch embeddings through multiple transformer layers with self-attention [4]. The output contains 384 channels which show both local texture details and complete anatomical information at a decreased spatial scale. The dense embeddings need percentile pooling to combine them into single feature vectors which function as compact representations for classification tasks.

The training data comes from SA-Med3D-140K which is a large collection of medical segmentation data that includes 143,000 3D masks which cover 245 anatomical categories and uses CT and MRI imaging modalities [16]. This scale of pretraining far exceeds what is typically available for individual tumor classification tasks and enables the model to learn generalizable representations of medical anatomy and pathology. Second, the model achieves efficient promptable segmentation that requires 10 to 100 times fewer prompt points than previous methods to produce satisfactory 3D segmentation results, indicating that the

learned representations capture clinically meaningful structure. Third, the SAM-Med3D-turbo variant used in this work was further fine-tuned on 44 additional datasets to improve cross-domain performance, making it particularly robust to variations in scanner protocols, image quality, and anatomical regions [16].

The choice of SAM-Med3D as the feature extraction backbone for this work is motivated by several considerations that align with the constraints and objectives of the classification task. The primary advantage is the ability to leverage rich pretrained representations without requiring large amounts of labeled tumor classification data. The process of training a volumetric encoder with similar characteristics from start would produce excessive model specialization because the SAM-Med3D encoder starts with a pre-trained model which achieves good results in different body areas and disease types. The model contains specific design elements which make it suitable for medical imaging because it handles the typical intensity ranges and noise patterns and anatomical features found in CT and MRI scans. The model differs from general-purpose vision models because it natively processes 3D volumetric data rather than analyzing individual slices, and was specifically trained on medical imaging rather than natural images. [12, ?].

Another practical consideration is computational efficiency. Pre-trained foundation models, including SAM-Med3D, can be used in a frozen configuration, meaning that feature extraction requires only a single forward pass through the network without gradient computation. This characteristic of foundation models dramatically reduces memory requirements and processing time compared to end-to-end training of a volumetric classification network. The extracted features become available for storage so they can be used in various classification experiments which allows researchers to test different model choices and hyperparameters and preprocessing methods at a fast pace without needing to perform the costly encoding process again. The experimental design of this work depends on separating feature extraction from classifier training because this approach enables researchers to evaluate transfer learning against end-to-end methods through controlled experiments.

3.3 Feature Standardization and Dimensionality Reduction

The third stage prepares the extracted feature vectors for classification through standardization and dimensionality reduction. Study-level embeddings produced by SAM-Med3D’s encoder have 384 channels and by default are pooled with percentile pooling (capturing five quantiles: 10th, 25th, 50th, 75th, 90th) to yield a 1,920-dimensional vector per case. The vectors undergo standardization to achieve zero mean and unit variance using training partition statistics, which prevent information leakage. Following standardization, PCA reduces the dimensionality while retaining the most informative variance. The standardization process ensures all feature dimensions contribute equally, preventing large numerical values from dominating the subsequent learning process.

The method of principal component analysis serves as an optional procedure which researchers should use when their pooling or aggregation approach

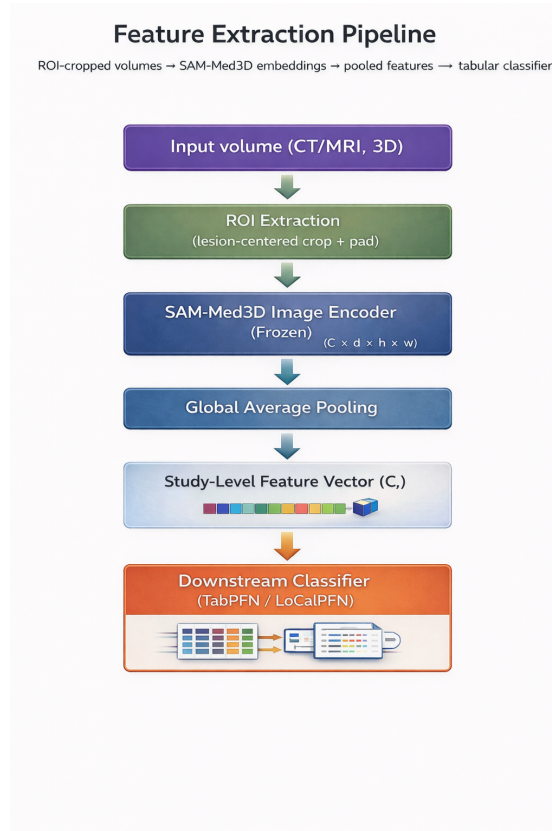


Fig. 3. Feature extraction pipeline using SAM-Med3D. Input volumes are preprocessed with ROI cropping, passed through the frozen image encoder, and aggregated via percentile pooling to produce feature vectors used for downstream classification.

generates output data that exceeds typical dimensions (this occurs when researchers combine different lesion embeddings through concatenation or when they replace average pooling with attention schemes or concatenation methods). When triggered, PCA is fitted on the standardized training features and the number of retained components is chosen as the minimum of a preset cap (default 500) and the numerical rank of the training feature matrix [8]. The default 384-dimensional average-pooled vectors disable PCA but the method becomes effective when different pooling techniques increase the data dimensions. The PCA transform which was trained on the data gets applied to both training and validation features to produce the final reduced representations.

Spatial pooling methods aggregate features within each lesion volume to create a fixed-size embedding. Three aggregation strategies are available—average pooling, multiscale pooling, and percentile pooling. Only average pooling produces a compact enough representation to avoid PCA dimensionality reduction;

the other strategies generate higher-dimensional vectors that require PCA before classification. The output of this stage is a set of normalized, optionally reduced feature vectors (one per case), with associated labels and identifiers, which serve as the input to the classification models.

3.4 TabPFN: Transformer-Based Classification for Small Tabular Datasets

The TabPFN (Tabular Prior-data Fitted Network), provides an application for a transformer based classifier which produces excellent performance with high speed on the smaller tabular classification problems. As it utilizes in-context learning, a single forward pass allows the classifier to generate predictions from training examples provided at inference time without updating parameters; i.e., the labeled examples are processed in one pass to allow the model to learn new task without using gradient-based training. For its part, TabPFN views supervised learning as a sequence modeling problem: It is trained prior to use on synthetic tabular datasets and then generates label predictions through the joint attention of both training examples (i.e., the "context") and test samples using a transformer-based architecture. At inference time, both the complete set of training data and the test sample are evaluated in a single computation eliminating the need to retrain the model and taking advantage of smaller datasets where retraining deep models can be expensive. Rather than fit a model to data, the data is provided as context to a pre-trained model that generalizes across a very broad distribution of synthetic classification tasks.

The architecture of TabPFN is based on a transformer encoder that processes both training and test samples simultaneously [6, ?]. During inference, the training set features and labels are concatenated with the test set features and passed through the transformer, which attends to all samples jointly. The output for each test sample is a probability distribution over the target classes, computed by attending to the training examples and implicitly learning decision boundaries in feature space. The design allows TabPFN to handle datasets with 10000 training examples and 500 features which it can process at a rate of one second on modern GPUs while outperforming other methods that require hyperparameter optimization and cross-validation.

The pretraining procedure of TabPFN is central to its effectiveness. The model was trained on millions of synthetic classification datasets generated from a carefully designed prior distribution that captures the structural properties of real-world tabular data [6]. This prior, called the Prior-data Fitted Network prior, samples datasets with varying numbers of features, class distributions, feature correlations, and nonlinear decision boundaries. By training on this diverse distribution of tasks, TabPFN learns a general-purpose classification algorithm that can be applied to new datasets without task-specific training. The pretraining objective is to minimize the cross-entropy loss on held-out samples from each synthetic dataset, effectively teaching the model to perform Bayesian posterior inference over the space of possible classifiers given the observed training data.

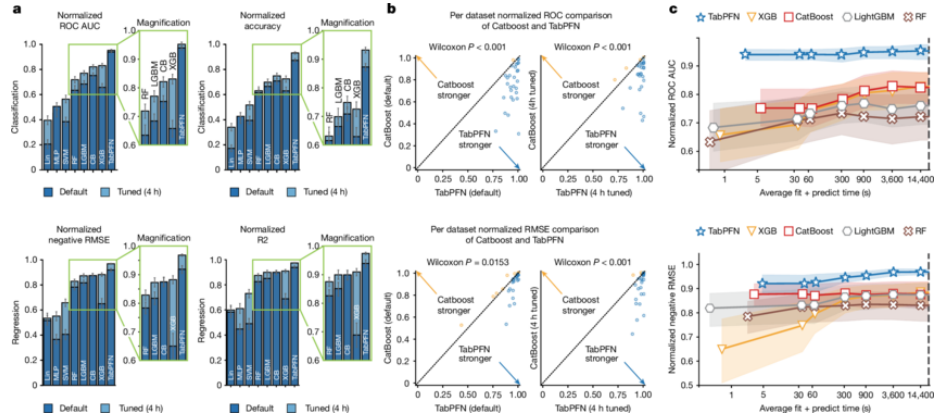


Fig. 4. Performance comparison of TabPFN against traditional machine learning methods. TabPFN achieves competitive accuracy while requiring no hyperparameter tuning and completing inference in significantly less time.

The new approach also pertains to the theoretical underpinnings. TabPFN can be seen as performing an approximate form of Bayesian inference over the posterior predictive distribution of labels based upon the training data [6, ?]. The pre-trained transformer provides a method to approximate the integral of all possible classifiers weighted by their posterior probability given the data (and the prior). It is this statistical basis which describes the success of TabPFN: in contrast to choosing a single hypothesis, TabPFN considers all hypotheses consistent with the data.

TabPFN was selected as the primary classification model for both practical and research reasons aligned with the task of identifying soft-tissue tumors. The advantage of TabPFN may be most pronounced when there are limited datasets. Medical imaging studies on rare tumors typically include a few hundred patients, as opposed to the thousands required for standard deep-learning classifiers to avoid overfitting; TabPFN is specifically designed to operate on datasets containing fewer than 10,000 examples. Therefore, the use of SAM-Med3D embeddings produces the mid-size vectors of features that TabPFN may process after reducing the dimensionality of these vectors, making TabPFN and SAM-Med3D compatible.

The TabPFN integration process occurs after the feature preparation stage in the classification pipeline. After SAM-Med3D embeddings are extracted and pooled to study-level vectors, the features are standardized to zero mean and unit variance using statistics computed on the training set. The analysis uses Principal component analysis to decrease data dimensions into 500 components which fulfills TabPFN feature restrictions and protects against high-dimensional space overfitting with insufficient sample data. The reduced features are passed to TabPFN along with binary labels, and class probabilities are obtained in a single forward pass. The TabPFN system enables users to make ensemble

predictions through different attention configurations which produce combined results for better stability but our initial tests revealed no performance gain for the evaluated datasets.

The model has a hard limit of 10000 training samples, which is not a constraint for the datasets in this study but would require alternative approaches for larger cohorts. Additionally, TabPFN was pretrained on synthetic data with specific distributional assumptions that may not perfectly match all real-world datasets. The thesis evaluates how different classifier selections affect result outcomes through a comparison with LoCalPFN which uses TabPFN to perform local neighborhood retrieval for better results on datasets that show better local than global patterns [15]. Native 3D classification heads trained end-to-end provide an additional baseline that does not rely on the TabPFN architecture.

3.5 LoCalPFN: Local Context Adaptation

Localizing feature space attention can improve performance when there are multiple populations of interest within a feature space, or when different parts of the feature space have different decision boundaries [1]. Medical imaging is one such domain, where tumors can have sub-types, and these sub-types typically reside in completely separate regions of the feature space, often with their own decision boundary [10]. Global approaches do not learn about these local class distribution features and relationships; they rely on global representations of all data points, which can lead to poor accuracy for some classes [15].

TabPFN’s approach to localizing attention to relevant subsets of the data involves two main components: 1) retrieving a neighborhood of data points near the input point of interest; 2) performing inference in TabPFN on this neighborhood [15]. The first component uses K-Nearest Neighbor (KNN) search in the Euclidean space of the standardized features to find the K nearest data points [2], whereas the second component uses TabPFN as described above.

The number of neighbors (K) is automatically determined as the smallest of 1000 and 10 times the square root of the total number of training examples [15]. For our dataset sizes (77 to 246 samples per category), we obtain approximately 88 to 157 neighbors, which allows us to retain locality and prevent excessive computations [15].

We also extend the KNN retrieval process with a lightweight adapter that can optionally be trained on TabPFN logits [7]. This adapter adjusts the output from the base model based on the information of the retrieved neighborhood, but does not modify the parameters of the TabPFN model. We train the adapter for a limited number of iterations with small learning rate to avoid overfitting to local biases in the data [15].

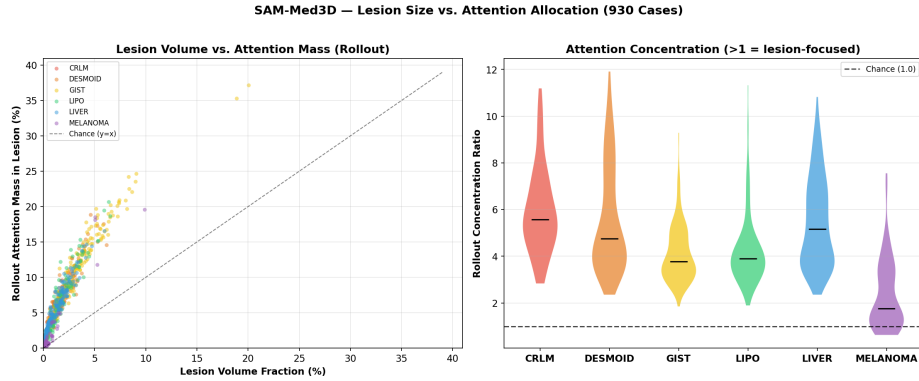


Fig. 5. SAM-Med3D attention allocation relative to lesion size across 930 cases from 6 datasets. *Left:* Scatter plot of lesion volume fraction vs. rollout attention mass within the lesion. Points above the dashed diagonal ($y = x$) indicate attention allocation exceeding chance level. *Right:* Violin plots of the rollout concentration ratio (attention mass / lesion volume fraction) per dataset. All datasets show median ratios well above 1.0 (dashed line), indicating that the pretrained encoder disproportionately attends to lesion regions without explicit lesion-specific supervision. CRLM exhibits the strongest concentration ($\sim 5.5\times$), while MELANOMA shows the weakest ($\sim 1.5\times$).

4 Experiments

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Table 2. Embedding diagnostic summary. Three tests probe SAM-Med3D feature quality at complementary levels: Mann-Whitney U (per-dimension), HSIC/CKA (global dependence), and k NN agreement (local geometry). Signal indicators: ● strong, ● weak, ● none.

Dataset	n	MW sig. dims	CKA	HSIC	p	k NN adj. ($k=5$)	k NN p
CRLM	77	0/1920 ●	0.069	0.0830	●	−0.046	0.6793 ●
Desmoid	203	395/1920 ●	0.052	0.0012	●	−0.037	0.7443 ●
GIST	246	679/1920 ●	0.080	0.0001	●	−0.008	0.5255 ●
Lipo	115	0/1920 ●	0.027	0.3245	●	0.113	0.0210 ●
Liver	186	0/1920 ●	0.023	0.1575	●	−0.082	0.9860 ●
Melanoma	103	0/1920 ●	0.033	0.3305	●	−0.037	0.6683 ●

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